

Breast Cancer Insights: Analytical Report



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Tools Used: Excel, Tableau

Project Type: Healthcare Data Analytics

Total Patients Analyzed: 334

Introduction:-

Breast cancer is one of the most commonly diagnosed cancers worldwide and remains a major public health concern. Understanding how patient demographics, tumor characteristics, and treatment choices influence disease progression and outcomes is critical for improving clinical decision-making. This project applies data analytics and statistical techniques to explore patterns and relationships within a breast cancer patient dataset.

Problem Statement:-

Breast cancer treatment and patient outcomes are influenced by a combination of demographic factors, tumor characteristics, biological markers, and treatment decisions. However, understanding how these variables interact is challenging without structured analysis.

The goal of this project is to analyze breast cancer patient data to identify key relationships between age, tumor stage, biological markers, surgery type, and follow-up outcomes. By applying exploratory analysis, statistical testing, and interactive visualization, the project aims to answer the following questions:

- 1. How does age vary across different tumor stages?**
- 2. Are tumor stages associated with demographic and biological factors such as gender, HER2 status, and protein expression?**
- 3. Do treatment choices (surgery type) influence follow-up duration?**
- 4. Which clinical factors show statistically significant associations, and which do not?**

The insights derived from this analysis are intended to support data-driven clinical understanding and demonstrate how analytics can be applied to healthcare datasets.

Data Description:-

This dataset contains clinical and demographic information of breast cancer patients who underwent surgical treatment for tumor removal. The data spans a short time frame and is suitable for exploratory analysis and hypothesis testing in a healthcare context. After data cleaning, the dataset consists of 334 patient records, with each row representing a unique patient.

The original dataset downloaded from **Kaggle**, contained 16 columns and more than 400 records. After removing incomplete and invalid entries, 18 columns 334 valid patient records were retained for analysis. Each record corresponds to an individual breast cancer patient.

- **Patient_ID** – Unique identifier for each patient
- **Age** – Age of the patient at diagnosis (in years)
- **Gender** – Male or Female
- **Protein1, Protein2, Protein3, Protein4** – Protein expression levels (units not specified)
- **Tumour_Stage** – Stage of cancer (I, II, III)
- **Histology** – Type of breast cancer (Infiltrating Ductal Carcinoma, Infiltrating Lobular Carcinoma, Mucinous Carcinoma)
- **ER Status** – Estrogen receptor status (Positive/Negative) (Only Positive in the dataset)
- **PR Status** – Progesterone receptor status (Positive/Negative) (Only Positive in the dataset)
- **HER2 Status** – Human Epidermal Growth Factor Receptor 2 status (Positive/Negative)
- **Surgery_Type** – Type of surgical procedure performed
- **Date_of_Surgery** – Date on which the surgery was performed
- **Date_of_Last_Visit** – Date of last recorded hospital visit (may be null)
- **Patient_Status** – Alive, Dead, or Unknown

Data Cleaning and Preprocessing:-

Data preparation was performed using **Excel Power Query** to ensure consistency and accuracy.

Cleaning Steps Performed:

- Standardized **Gender values** to remove inconsistencies
- Removed **blank rows** with no patient information
- Reduced dataset to **334 valid patient records**
- Retained **null values** in *Date of Last Visit* for patients who did not return after surgery
- Replaced null values in **Patient Status** with “**Unknown**”
- Calculated **Follow-up Days** using:
 - Date of Last Visit – Date of Surgery
- Created **Age Groups** for categorical analysis:
 1. 29–39
 2. 40–49
 3. 50–59
 4. 60–69
 5. 70+

Exploratory Data Analysis (EDA):-

I24												
	A	B	C	D	E	F	G	H	I	J	K	L
1	Tumor_Stage Count			Tumor_Stage+Gender					Tumor_Stage+Patient_Status			
2	Tumor_Stage	Patient_Count		Count of Patient_ID	Gender				Count of Patient_ID	Person_Status		
3	I	64		Tumor_Stage	Female	Male	Grand Total		Tumor_Stage	Alive	Dead	Unknown
4	II	189		I	64		64		I	51	10	3
5	III	81		II	186	3	189		II	144	38	7
6	Grand Total	334		III	80	1	81		III	60	18	3
7				Grand Total	330	4	334		Grand Total	255	66	13
8												
9												
10	HER2_Status+Tumor_Stage							Patient_Status+Surgery_Type				
11	Count of Patient_ID	Tumor_Stage						Count of Patient_ID	Surgery_Type			
12	HER2_Status	I	II	III	Grand Total			Patient_Status	Lumpectom	Modified Radical	Mastectomy	Other
13	Negative	61	176		68	305		Alive	57	72	73	53
14	Positive	3	13		13	29		Dead	9	20	25	12
15	Grand Total	64	189		81	334		Unknown		4	7	2
16								Grand Total	66	96	105	67
17	Surgery_Type+Age_Group											
18	Count of Patient_ID	Age_Group										
19	Surgery_Type	29-39	40-49	50-59	60-69	70+	Grand Total					
20	Lumpectomy	3	10		15	19	19					
21	Modified Radical Mastectomy	5	24		34	18	15					
22	Other	3	22		33	25	22					
23	Simple Mastectomy	4	15		16	17	15					
24	Grand Total	15	71		98	79	71					

Fig1. EDA: Excel Pivot Tables

EDA was conducted using **Excel Pivot Tables** to understand patient distribution and clinical patterns.

Key Observations:

- **Tumor Stage Distribution:**
 - Stage I: 64 patients
 - Stage II: 189 patients
 - Stage III: 81 patients
- Initial exploration revealed:
 1. High prevalence of **female patients**
 2. Majority of patients clustered in **Tumor Stage II**
 3. Noticeable variation in surgery types and outcomes

EDA helped identify meaningful variables for **statistical testing and dashboard analysis**.

Statistical Analysis and Hypothesis Testing:-

Hypothesis testing was conducted in **Excel**, supported by **VBA macros** to dynamically populate a **Hypothesis summary results table**.

Hypothesis Testing Summary							
Sr.No.	Test	Variables	Test Used	p-value	Decision	Conclusion	Click to see the tests
1	Age vs Tumor_Stage	Age,Tumor_Stage	ANOVA	0.008	Reject Ho	Mean Age differs across Tumor_Stages	See the test
2	Gender vs Tumor_Stage	Gender,Tumor_Stage	Chi-square	0.6	Accept Ho	No association between Gender and Tumor_Stages	See the test
3	Histology vs Tumor_Stage	Histology,Tumor_Stage	Chi-square	0.42	Accpet Ho	No association between Histology and Tumor_Stages	See the test
4	HER2_Status vs Tumor_Stage	HER2_Status,Tumor_Stage	Chi-square	0.02	Reject Ho	Significant association between HER2_Status and Tumor_Status	See the test
5	Protiens vs Tumor_Stage	Proteins1-4,Tumor_Stage	ANOVA	0.49-0.77	Accept Ho	Protein markers are not associated with Tumor_Stages	See the test
6	Proteins vs Histology	Proteins1-4,Histology	ANOVA	0.36-0.68	Accept Ho	Protein markers are not associated with Histology	See the test
7	Proteins vs HER2_Stautus	Proteins1-4,HER2_Status	ANOVA	0.58-0.95	Accept Ho	Protein markers are not associated with HER2_Status	See the test
8	Proteins vs Surgery_Type	Proteins1-4,Surgery_Type	ANOVA	0.015-0.12	Reject Ho	Protein1 showed statistically significant differnece across Surgery_Type whereas Protein2,3,4 did not show the variance	See the test
9	Tumor_Stage vs Patient_Status	Tumor_Stage,Patient_Status	Chi-square	0.61	Accept Ho	No association between Tumor_Stages and Pateint_Status	See the test
10	Histology vs Pateint_Status	Histology,Patient_Status	Chi-square	0.88	Accpet Ho	No association between Histology and Patient_Status	See the test
11	Surgery_Type vs Followup_Days	Surgery_Type,Followup_Days	ANOVA	0.003	Reject Ho	Significant association between Followup_Days and Surgery_Type	See the test

Fig2. Hypothesis Testing Summary

ANOVA was used for numerical vs categorical variables, and **Chi-square tests** were used for categorical variable comparisons.

1. **Age VS Tumor Stage (ANOVA):**

Age is significantly associated with tumor stage, indicating that tumor progression varies across age groups.
2. **Gender VS Tumor Stage (Chi-square):**

No significant association was found, meaning tumor stage distribution is independent of gender and is not limited to any specific gender.
3. **Histology VS Tumor Stage (Chi-square):**

No significant association observed; cancer type does not determine tumor stage at diagnosis in this dataset.
4. **HER2 Status VS Tumor Stage (Chi-square):**

A significant association exists, suggesting HER2 status is linked to tumor progression.
5. **Protein Expression VS Surgery Type (ANOVA):**

Protein1 shows a significant association with surgery type, while Protein2, Protein3, and Protein4 do not.
6. **Surgery Type VS Follow-up Days (ANOVA):**

Follow-up duration varies significantly by surgery type, reflecting differences in post-surgical care requirements.
7. **Tumor Stage VS Patient Status (Chi-square):**

No significant association found; patient outcome cannot be inferred solely from tumor stage.
8. **Histology VS Patient Status (Chi-square):**

No significant association observed between cancer type and patient status.

Dashboard Insights:-

- **Total Patients:** 334
- **Average Age:** 58.89 years
- **Gender Distribution:** 98.80% Female, 1.20% Male i.e.~1 in 8 women diagnosed with breast cancer while ~1 in 1000 men diagnosed
- **Patient Status:** 76.35% Alive, 19.76% Dead, 3.89% Unknown
- **Most Common Cancer Type:** Infiltrating Ductal Carcinoma

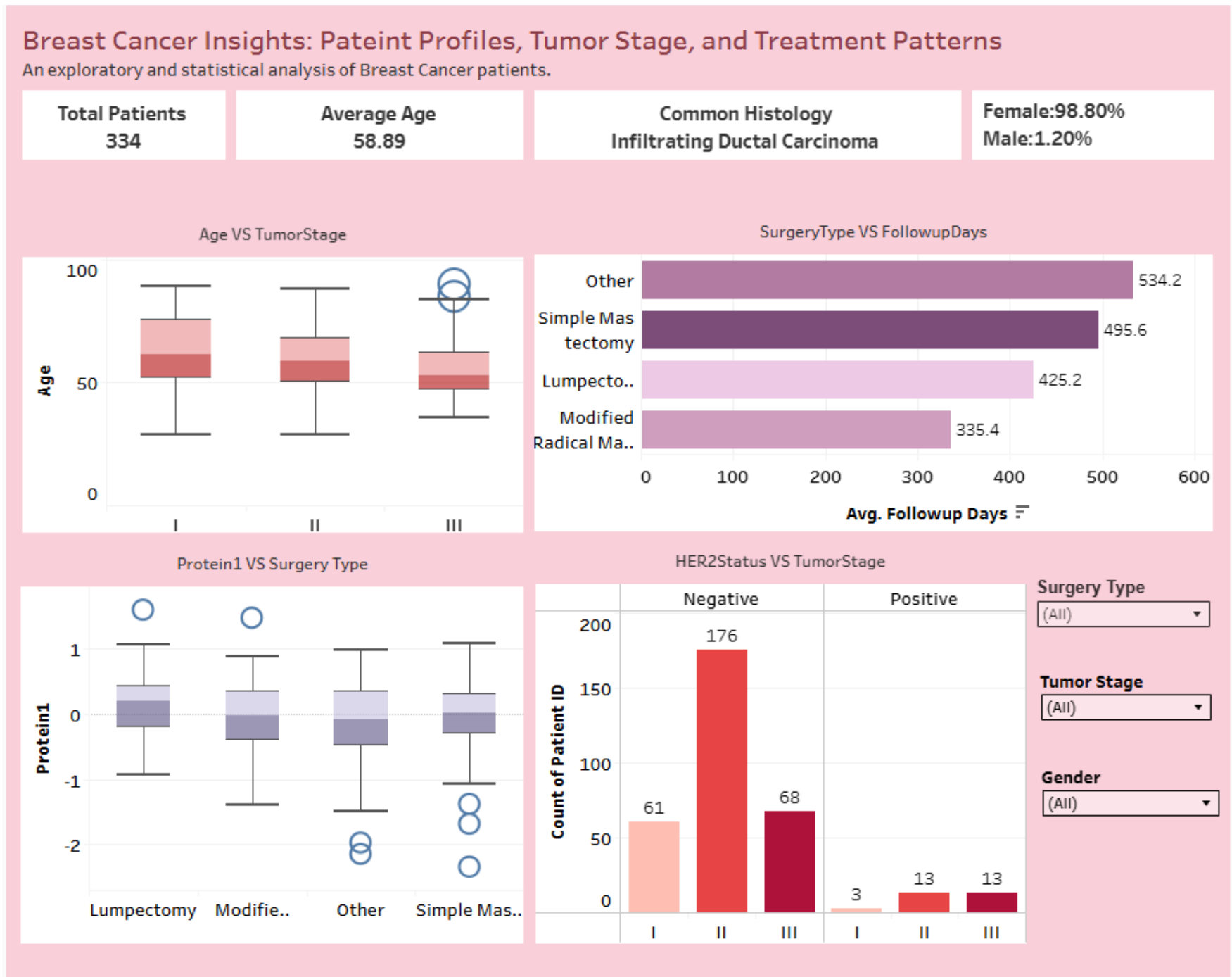


Fig3. Breast Cancer Insights Dashboard

- **Age VS Tumor Stage:**
Median age decreases with tumor severity — **Stage I: 61.5 yrs, Stage II: 59 yrs, Stage III: 53 yrs**, indicating earlier onset in advanced stages.
- **Protein1 VS Surgery Type:**
Median Protein1 is highest for **Lumpectomy (0.199)** and lowest for **Other surgeries (-0.080)**, confirming Protein1’s association with surgical choice.
- **Surgery Type VS Follow-up Days:**
Other surgery types show the longest average follow-up duration, followed by **Simple Mastectomy, Lumpectomy**, and **Modified Radical Mastectomy**.
- **HER2 Status VS Tumor Stage:**
HER2-negative Stage II patients form the largest group (**176 patients**), while HER2-positive cases remain low across all stages (**≤13 patients per stage**).
- **Cancer Histology:**
Infiltrating Ductal Carcinoma dominates all tumor stages, while **Mucinous Carcinoma** is rare and limited to early stages.

Breast Cancer Insights: Pateint Profiles, Tumor Stage, and Treatment Patterns

An exploratory and statistical analysis of Breast Cancer patients.



Fig4. Breast Cancer Insights Dashboard (Gender: Male)

Key Interpretations:-

- Tumor severity varies across age groups, with advanced stages appearing at lower median ages.
- Biological markers (HER2 status and Protein1 expression) show stronger relationships with tumor characteristics and treatment patterns than demographic variables.
- Gender does not independently influence tumor stage distribution, reinforcing that breast cancer severity is not gender-dependent within this dataset.
- Surgical approach is linked to post-operative follow-up duration, indicating differing care pathways.
- Rare histological subtypes are limited to early tumor stages and represent a small proportion of cases.

Conclusion:-

This project demonstrates an **end-to-end data analytics workflow**, combining data cleaning, exploratory analysis, statistical testing, and interactive visualization. The findings reveal meaningful relationships between **demographics, tumor stage, biological markers, treatment types, and follow-up outcomes**, while also highlighting variables with no statistical association.